

Krittin Jindamai (Yim)

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EDUCATION

University of Florida

Bachelor of Science in Aerospace Engineering
Davis UWC Scholar | Honors Program | Dean's List

Expected Graduation: May 2027

GPA: 3.98 / 4.00

RESEARCH EXPERIENCE

Nonlinear Control Research Group, Gainesville, Florida

Undergraduate Research Assistant

Spring 2026 - Present

- Investigate Control Barrier Functions (CBFs) for smooth waypoint tracking of nonholonomic robots near safety constraints.
- Simulate multi-robot and drone systems in ROS2 and Gazebo to evaluate CBF-based safety control before deployment on physical robots.

Space Systems Group, Gainesville, Florida

Undergraduate Honors Thesis

Summer 2026 - Present

- Investigate singularity escape and avoidance algorithms for Hybrid Control Moment Gyroscopes (HCMGs).
- Develop a 12U CubeSat simulation in sun-synchronous orbit for rapid retargeting and precision nadir pointing under gravity-gradient, aerodynamic-drag, and J2 perturbations.
- Evaluate Magnetorquer (MTQ)-based momentum dumping for HCMG desaturation.

WORK EXPERIENCE

Mechanics of Materials Laboratory, Gainesville, Florida

Teaching Assistant

Fall 2026 - Present

- Lead two laboratory sections of 60+ students conducting experiments with dynamic data acquisition (DAQ), Wheatstone bridges, strain gauges, and Instron machines.
- Hold office hours and assist students with laboratory procedures, course material, and technical report writing.

Geo-Informatics and Space Technology Development Agency, Thailand

Mechanical Engineering Intern

Summer 2025

- Meshed and assembled the primary structure of Thailand's THEOS-3A using HyperMesh.
- Performed free-free and fixed-base modal analyses to determine natural frequencies and mode shapes of the satellite structure.
- Performed quasi-static load analyses under multiple load factors using MSC NASTRAN to evaluate structural integrity during launch and on-orbit disturbances.

Swamp Launch Rocket Team, Gainesville, Florida

Controls Lead Engineer

Fall 2025 - Spring 2026

- Developed embedded C++ software for real-time trajectory prediction and active rocket altitude control using PID control.
- Designed and manufactured a custom flight-computer PCB and CAM-slot airbrake mechanism using CNC machining, milling, and turning.
- Implemented an Extended Kalman Filter to estimate vehicle position and orientation in real time.

Software Associate Lead Engineer

Fall 2024 - Spring 2025

- Led a 15-member software sub-team to a 6th-place finish at the NASA 2025 Student Launch Competition.
- Implemented a Kalman Filter to fuse IMU and barometer measurements for real-time rocket velocity estimation.
- Developed a C++ program to collect and transmit rocket and landing-site data to a NASA transceiver over 2M radio.
- Collaborated with electrical and mechanical teams to develop and test the rocket retention mechanism.

SKILLS AND CERTIFICATIONS

Technical Skills: MATLAB | Simulink | ROS2 | Gazebo Harmonic | Python | C++ | Ansys | HyperMesh | MSC NASTRAN | Solidworks (CSWA Certified) | Fusion360 | Sine and Shock Testing (VRU Certified)